

MD MINHAZUR RAHMAN

Chittagong, Bangladesh

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| [Live Research Demo](#)

ACADEMIC PROFILE

Data Science researcher and educator with an MSc from the University of Greenwich (UK) and over two years of enterprise IT experience at GPH Ispat Limited, one of Bangladesh's largest publicly listed industrial corporations. Specialising in **privacy-preserving machine learning**, **synthetic data generation**, and **applied time-series forecasting**, with a DOI-indexed preprint on Zenodo and a publicly deployed interactive research dashboard. Committed to advancing data science education and applied AI research in Bangladesh through rigorous, research-informed teaching and reproducible, open-source scholarship.

EDUCATION

University of Greenwich

London, United Kingdom

MSc Data Science — Pass with Merit (180 Credits, Level 7)

Sep 2024 – Jan 2026

- **Dissertation (70%):** "Real-Time Retail Demand Forecasting Using Synthetic Data: A Privacy-Preserving SKU-Level Pipeline." Published as a first-author DOI-indexed preprint on Zenodo with a live deployed Streamlit dashboard.
- **Strong Module Results:** Applied Machine Learning (71%), Cloud Computing & HPC (74%).
- **Key Coursework:** Statistical Methods for Time Series, Big Data Analytics, Data Visualisation, Machine Learning, High-Performance Computing.

Independent University, Chittagong

Chittagong, Bangladesh

BSc in Computer Science & Engineering

Jan 2017 – Jan 2021

- **Final Year Project:** "Rare Event Detection in Crime Videos Using 3D-ResNet18 and Custom-CNN" — designed and implemented deep learning models for rare event detection in surveillance video using PyTorch; applied spatio-temporal feature extraction on the UCF Crime Dataset; conducted comparative model performance analysis on imbalanced real-world video classification data.

PUBLICATIONS & PREPRINTS

- **Rahman, M. M.** (2026). "Synthetic Retail Time-Series for Privacy-Preserving Demand Forecasting: An End-to-End Pipeline, Benchmark Evaluation, and Privacy Analysis." *Preprint — Zenodo*. DOI: [10.5281/zenodo.19479285](https://doi.org/10.5281/zenodo.19479285).
Interactive Demo: retail-forecasting.streamlit.app | *Google Scholar:* scholar.google.com | *ORCID:* [0009-0008-9418-6614](https://orcid.org/0009-0008-9418-6614)

RESEARCH EXPERIENCE

MSc Dissertation Research

University of Greenwich, London

Privacy-Preserving Synthetic Data for Retail Demand Forecasting

2024 – 2025

- Designed a controllable parametric generator for synthetic retail time-series capturing trend, seasonality, promotions, holidays, and exogenous traffic — eliminating real-world privacy risks entirely.
- Engineered leakage-safe features (rolling windows, lag variables) with strict rolling-origin cross-validation (ROCV) to prevent temporal data leakage.
- Benchmarked Random Forest and Gradient Boosted Trees (LightGBM) against linear baselines using scikit-learn pipelines, achieving a **37% reduction in MAE** and **11.7% MAPE** on short-horizon forecasts.
- Deployed an interactive [Streamlit dashboard](#) enabling real-time SKU selection, horizon control, and promotion/holiday scenario analysis.
- Evaluated synthetic data quality using a Fidelity–Utility–Privacy (FUP) framework; published results as a first-author DOI-indexed preprint on Zenodo.

HPC & Parallel Computing Research

University of Greenwich, London

Coursework Project — Clouds, Grids & Virtualisation

2025

- Developed and optimised a parallel 2D Jacobi numerical solver in C++ using OpenMP on the university HPC cluster running on Microsoft Azure infrastructure.
- Achieved a **4.81× speedup** on 8 threads; profiled memory-bound behaviour, cache effects, and strong-vs-weak scaling under distributed cloud-connected HPC environments.

TEACHING & MENTORING

Voluntary Teaching Support

MSc Data Science Programme

University of Greenwich, London

Jan 2025 – Sep 2025

- Provided supplementary instruction to 10+ MSc peers in Python-based ML workflows, covering scikit-learn pipelines, feature engineering, and model evaluation.
- Supported fellow students with debugging, conceptual clarification, and coursework strategy across Machine Learning and Big Data modules.
- Mentored peers through dissertation planning and methodology, with particular focus on experimental design and reproducible research practice.

Independent Academic Tutor

Mathematics, English & Python (part-time, evenings)

Chittagong, Bangladesh

Jan 2020 – Aug 2024

- Delivered structured private tuition to 15+ students (ages 14–22) in mathematics, English, and Python programming across four years, maintained consistently alongside full-time employment.
- Majority of students demonstrated measurable improvement in subject performance following structured one-to-one intervention; multiple students progressed from failing to passing grade boundaries in core subjects.
- Several students subsequently secured university admissions abroad, including institutions in the United Kingdom, Malaysia, and Germany.
- Designed personalised learning plans adapting to individual needs across secondary and pre-university levels, with emphasis on step-by-step problem decomposition and rebuilding understanding from first principles.

INDUSTRY EXPERIENCE

GPH Ispat Limited

Officer — Information Technology Department

Chittagong, Bangladesh

Feb 2022 – Aug 2024

- Served as IT Officer at one of Bangladesh's largest publicly listed steel manufacturing corporations (Dhaka Stock Exchange listed), supporting enterprise-scale software and infrastructure operations.
- Administered and maintained internal software systems accessed daily by staff across departments, ensuring operational continuity of IT infrastructure.
- Monitored and reviewed inventory tracking data, supporting data integrity across materials management processes.
- Maintained and troubleshooted enterprise systems and internal workflows, reducing operational disruptions through proactive IT support and process management.
- Developed internal tools and automation scripts using Python and SQL to support departmental operations and reporting workflows.

Vcube Soft & Tech

QA Engineer — Computer Vision Data

Chittagong, Bangladesh

2021 – Feb 2022

- Curated and validated annotated datasets from 360° imagery for production-grade 3D floor-plan reconstruction ML systems.
- Enforced metadata consistency and implemented QA validation pipelines, directly contributing to training data quality for deployed computer vision models.

OPEN-SOURCE SOFTWARE & DEPLOYED SYSTEMS

- **Synthetic Retail Forecasting Pipeline** | [GitHub](#) | [Live Demo](#)
End-to-end Python codebase: parametric time-series generation, feature engineering, scikit-learn and LightGBM ensemble training, and Streamlit deployment. Evaluated via the Fidelity–Utility–Privacy (FUP) framework. DOI-indexed preprint published on Zenodo.
- **Data Science Teaching Portfolio** | [GitHub](#)
Structured teaching notebooks for undergraduate data science education in Bangladesh, covering time-series forecasting, feature engineering, and applied machine learning with tree-based models. Designed to bridge research-grade methods and classroom accessibility.
- **HPC Parallel Computing Benchmarks** | [GitHub](#)
Reproducible C++ OpenMP benchmarking suite for shared-memory parallelism; includes Python multiprocessing experiments for scaling ML workloads on HPC infrastructure.

TECHNICAL SKILLS

- **Machine Learning & Data Science:** scikit-learn · LightGBM · Random Forest · XGBoost · Time-Series Forecasting · Synthetic Data Generation · Feature Engineering · Privacy-Preserving ML
- **Deep Learning:** PyTorch · 3D-ResNet18 · Custom CNN Architectures · Spatio-Temporal Feature Extraction · Imbalanced Classification
- **Cloud & HPC:** Microsoft Azure · HPC Cluster Computing · OpenMP · MPI · Distributed Computing Environments
- **Big Data:** Hadoop · Large-Scale Data Processing Pipelines
- **Programming & Data:** Python · SQL · C++ · R · Pandas · NumPy · Matplotlib · Seaborn · Jupyter Notebook
- **Deployment & Developer Tools:** Streamlit · Git · GitHub · LaTeX
- **Research Methods:** Rolling-Origin Cross-Validation · Experimental Design · Statistical Diagnostics · Reproducible Research · Benchmark Evaluation
- **Enterprise IT:** Software Development · IT Infrastructure Management · Systems Administration · Inventory Data Management

COURSES PREPARED TO TEACH

Introduction to Data Science · Machine Learning · Applied Statistics & Time-Series Analysis · Python Programming · Big Data Analytics · Research Methods in Computing · Data Visualisation · Introduction to Artificial Intelligence · Software Engineering Fundamentals

SCHOLARLY PROFILES

ORCID: orcid.org/0009-0008-9418-6614 | **Google Scholar:** scholar.google.com/citations?user=CQOSSHIAAAAJ
| **GitHub:** github.com/minhazda | **LinkedIn:** linkedin.com/in/mohammadminhaz

HONORS & AWARDS

Winner — Game of Presentations <i>Inter-University Technical Presentation Competition</i>	2019
Second Runner-Up — Bankers' Hunt Bangladesh <i>City Bank National Case Competition</i>	2020

REFERENCES

Dr. Tuan Vuong | MSc Dissertation Supervisor | University of Greenwich, London | T.Vuong@greenwich.ac.uk
Additional academic and professional references available upon request.